

model definition

Types of Loss function:

Regression:

- mse
- mae
- huber loss

Classification:

- Binary Cross Entropy
- Categorical Cross Entropy
- hinge loss

Autoencoders:

kL divergence

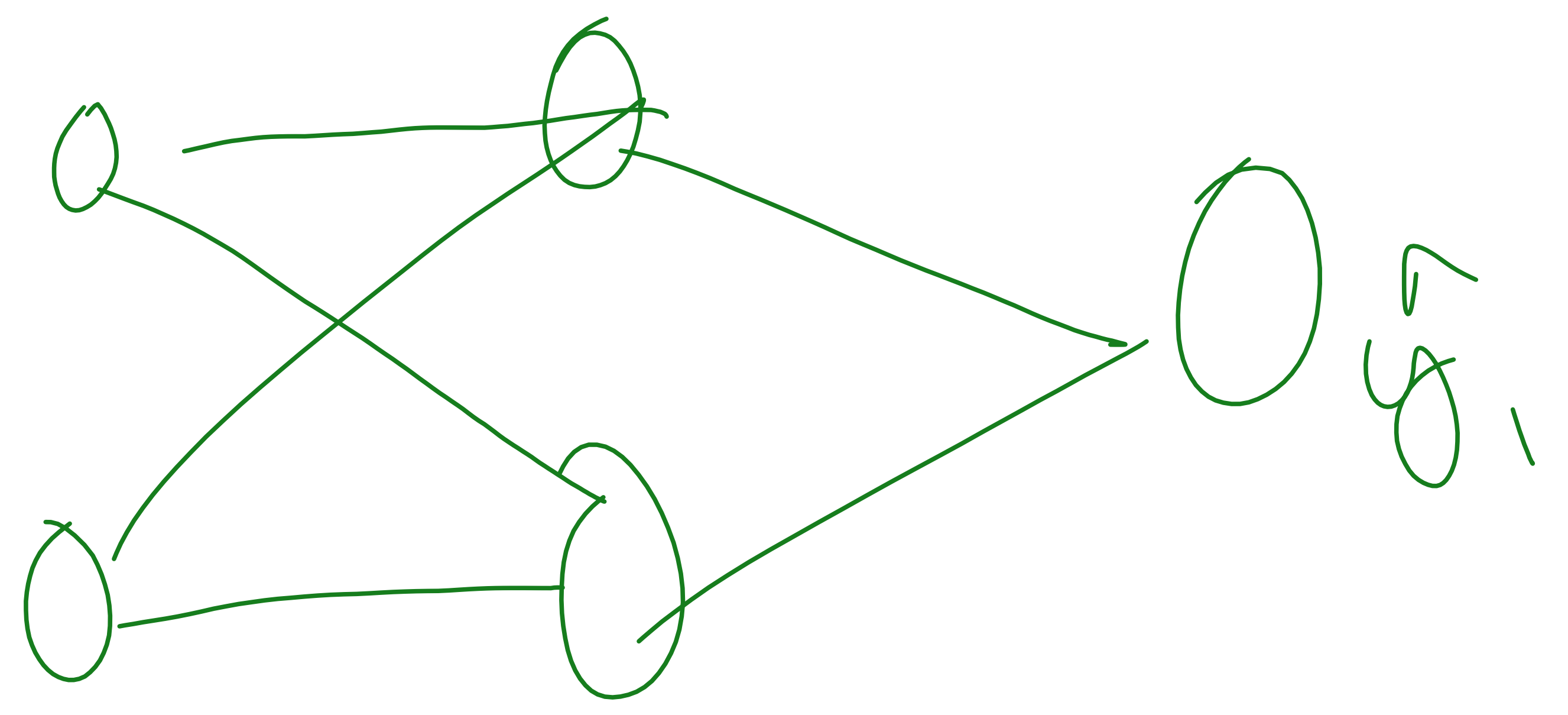
GAN

- discrimination loss
- minimax gan loss

Loss functions

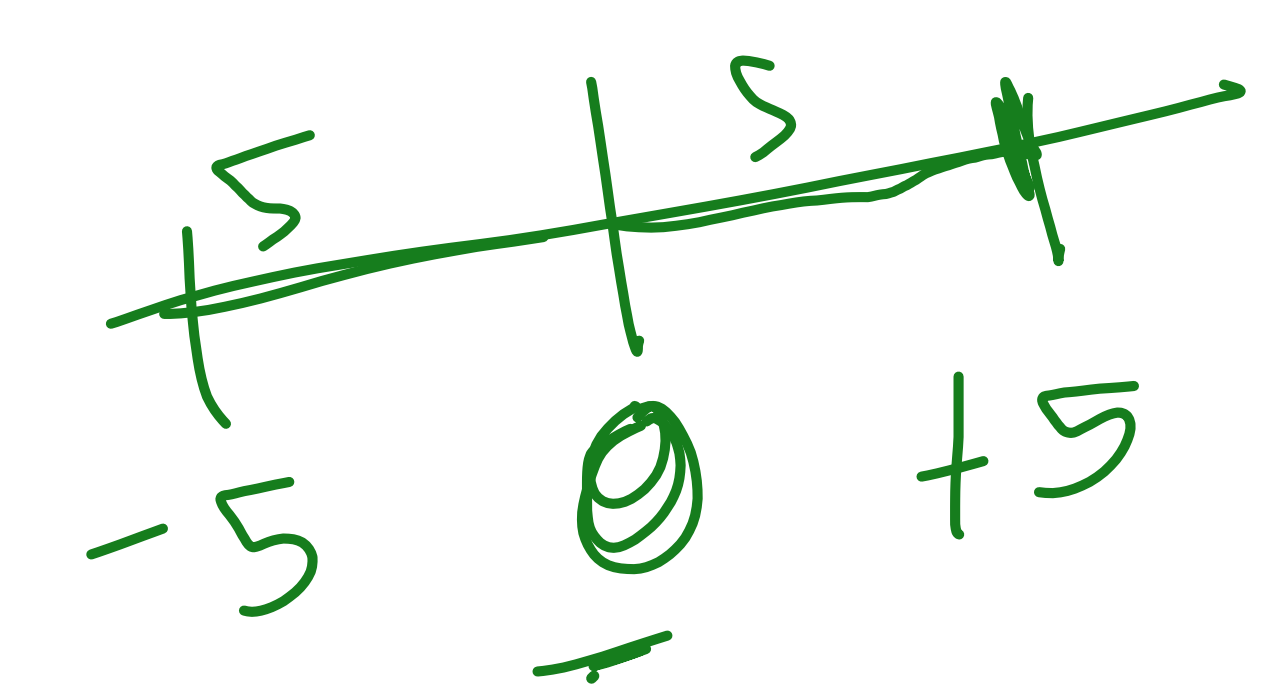
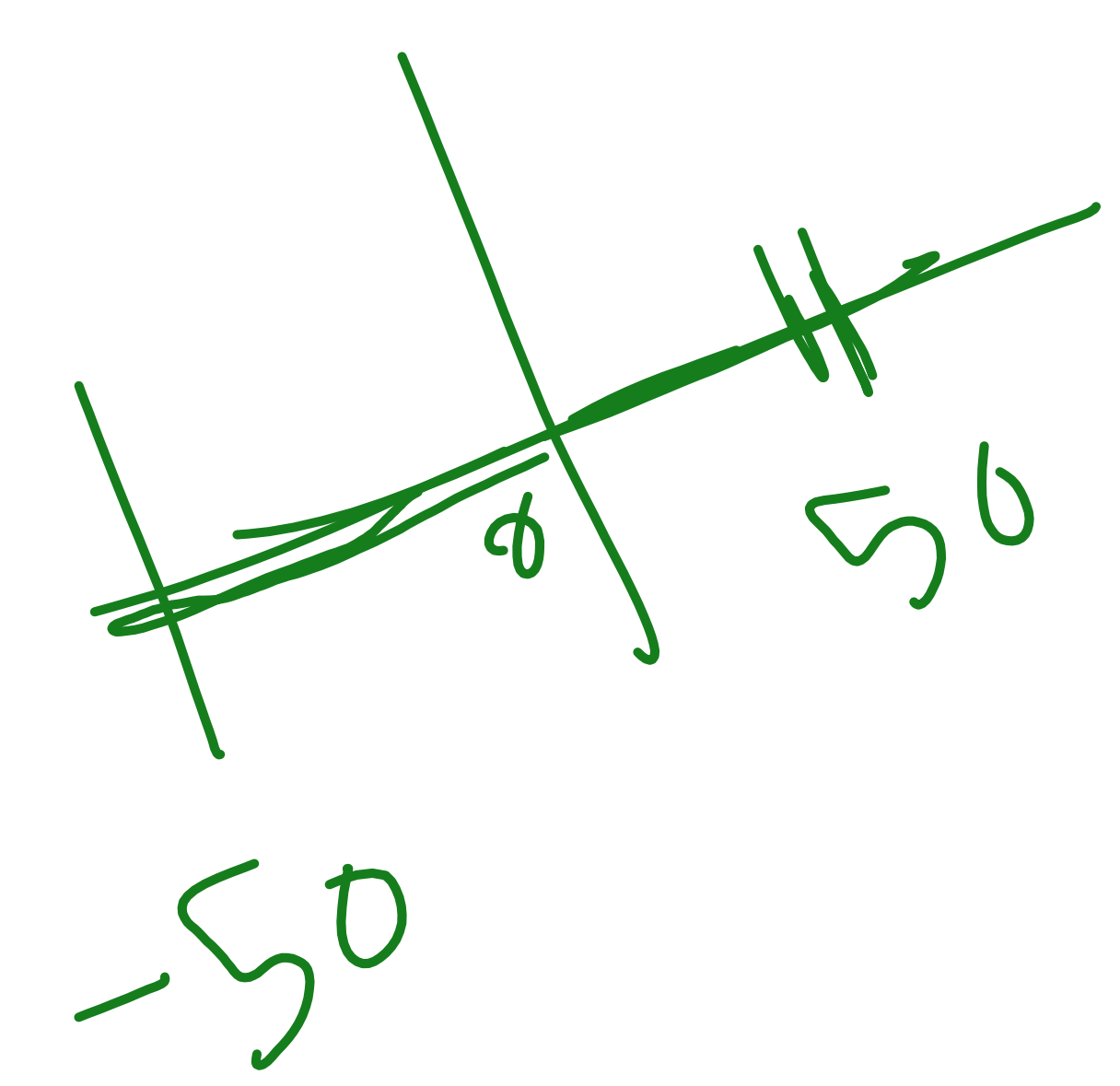
Study hours
2

Sleep



dataset

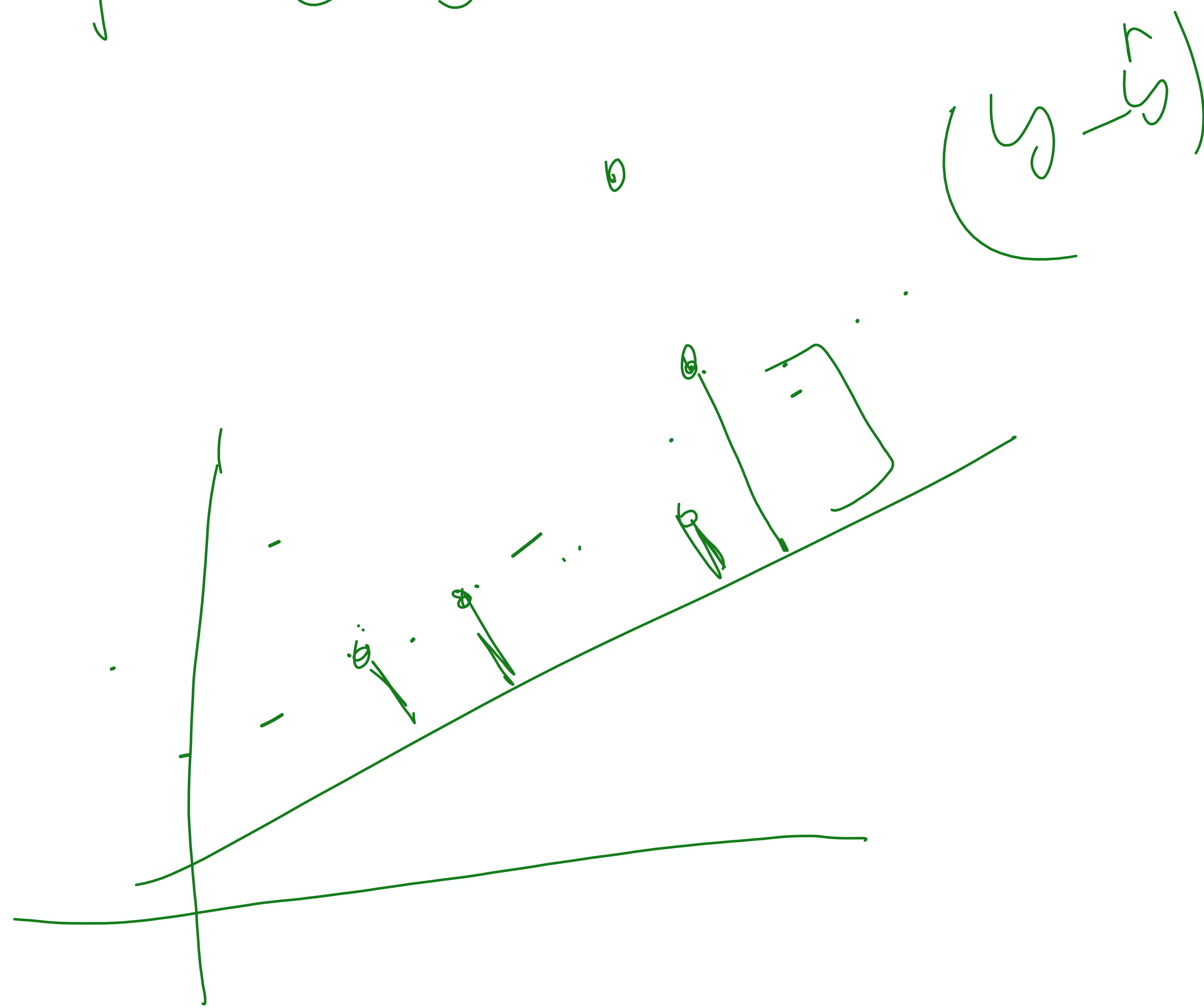
Study hours	Sleep	Attendance	marks
2	3	5	100
6	7	8	200
9	10	11	300



-50
50

Mean Square Error (MSE)

$$(y_i - \hat{y}_i)^2$$



$$\begin{array}{r} 2-3-4 \\ 4-9-16 \end{array}$$

Advantages:

$$\frac{(n-1)^2}{2}$$

- 1) Easy to interpret
- 2) Diffiable $\rightarrow$ ap

3) 1 local minima

Disadvantages

✓ 1) robust to outliers (not)

2) required

Linear

Activation fn

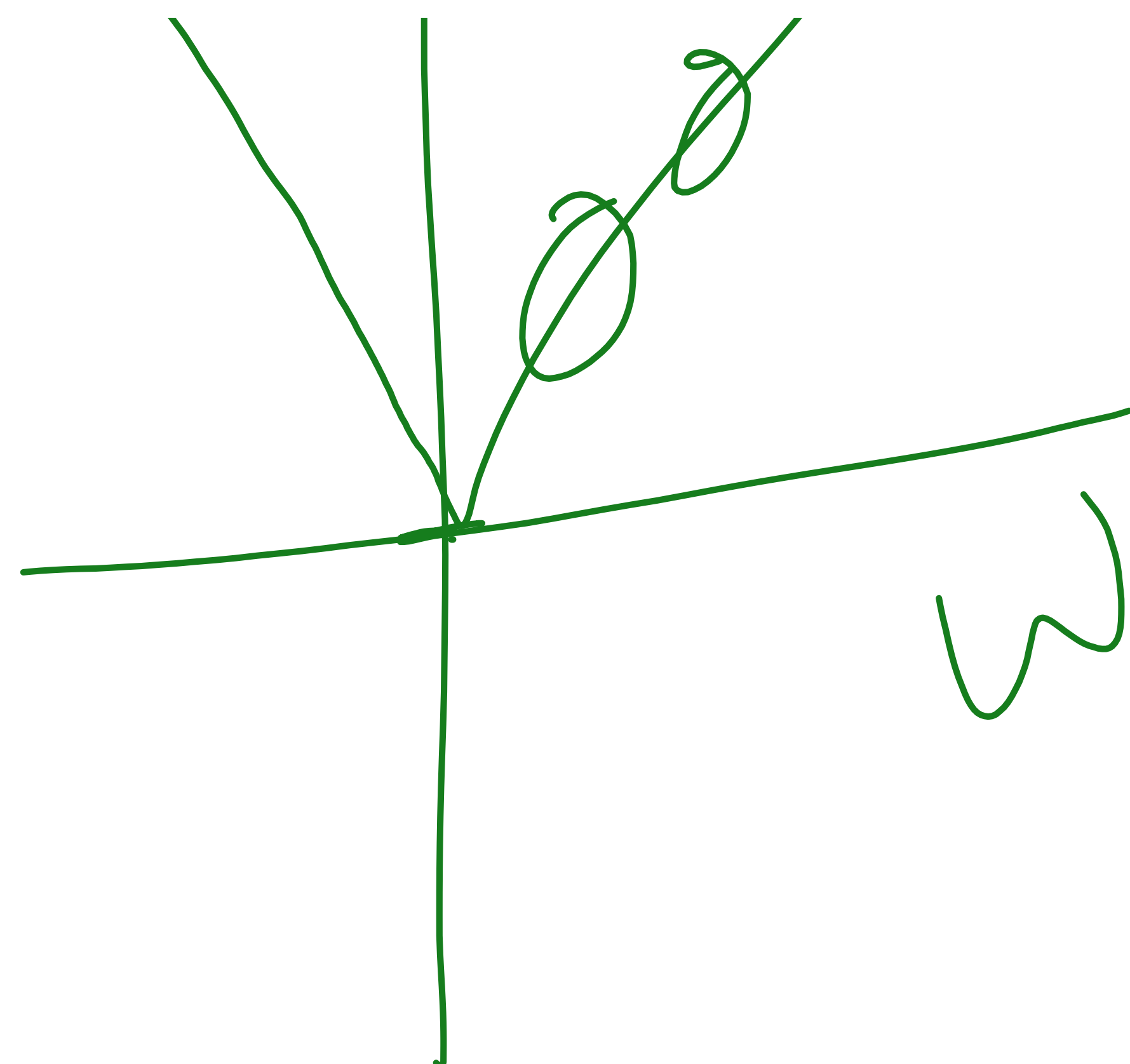
Q26 ✓

Mean Absolute Error: mae

$$Loss = |y_i - \hat{y}_i|$$

2d 1) Interpret
re visit outliers ✓

- 2) 0000
- 3) Robust to



disadv

Computationally expensive

Huber loss

$\rightarrow$ MSE

error

1 1 1 1 2 1 1

$$L(w, \hat{y}) = \begin{cases} \frac{1}{2} (y - \hat{y}) & \text{if } |y| \\ \delta |y - \hat{y}| - \frac{1}{2}\delta & \text{if } |y| \end{cases}$$

$$y = [5, 8, 3]$$

$$\hat{y} = [5.7, 7.8, 1]$$

$$|(5 - 5.7)| = 0.7$$

Adv
1) Robust to outliers
2) Faster than
MSE

Disadv
1) Requires
tuning
2) Slightly
complex

Classification

1 0

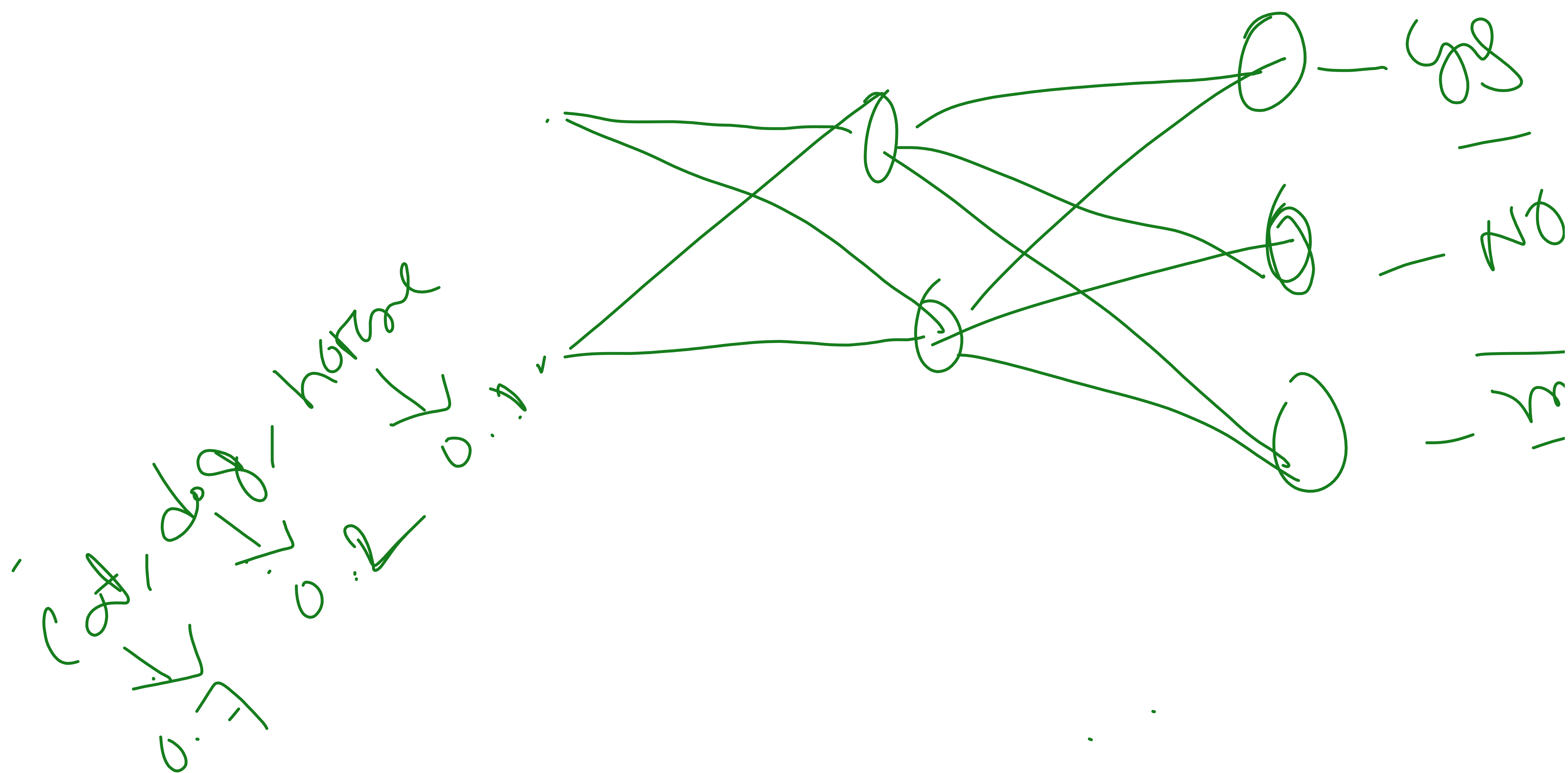
one-hot

dataset

Study hours	Sleep	Attendance	Pass?	yes
2	3	5	yes	1
6	7	8	NO	0
21	10	41	maybe	0

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

$$L = -y_1 \log(\hat{y}_1) - y_2 \log(\hat{y}_2) - y_3 \log(\hat{y}_3)$$



Adv:

1) works naturally with
probabilities
very effective for
non-stationary

4) multi-class
Classification

Disadvantage:

- 1) sensitive for
mislabelled data.
- 2) Class imbalance (ser

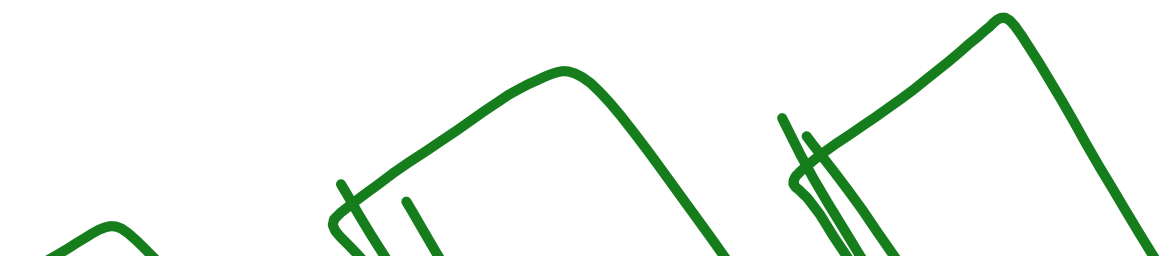
✓ Main
 formula: $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

$$\frac{\partial L}{\partial w}$$

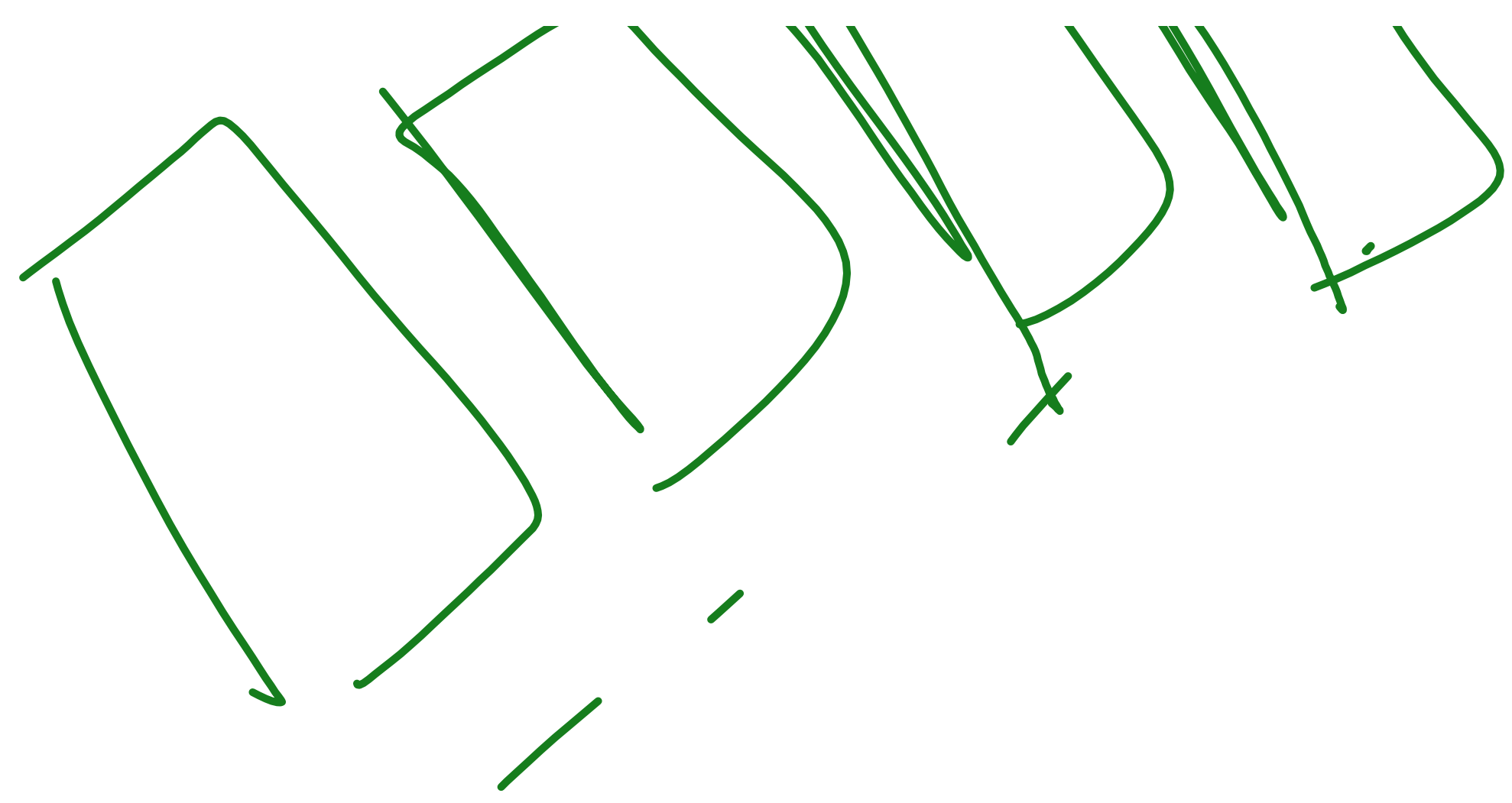
$$\frac{\partial L}{\partial w}$$

vanishing Gradient Problem

± 0.1



$$\frac{0.000}{0.7 + 0.7 + 0.1 + 0.7}$$



row 5 Anime loop

- 1) Deep NN
- 2) Sigmoid / tanh

How to get rid of vanishing
gradient prob

1) Reduce complexity

2) ReLU

3) batch normalization

* BCF
* hinge loss